

COLD BAY, AK, USA

WMO: 703160

Lat: 55.221N

Lon: 162.732W

Elev: 24

StdP: 101.04

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25624

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-10.9	-15.7	1.0	-12.3	-13.9	1.1	-10.4	20.6	2.9	18.3	1.3	7.0	340	0.955

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.5	16.1	13.2	14.8	12.4	13.8	11.8	13.8	15.5	12.9	14.2	12.2	13.3	6.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.9	9.3	14.3	12.3	8.9	13.6	11.6	8.5	12.9	38.7	15.5	36.6	14.2	34.8	13.3	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.6	14.3	DB	-15.6	19.3	3.4	2.0	-18.0	20.7	-20.0	21.9	-21.9	23.0	-24.4	24.4	(4)
(5)			WB	-16.0	15.6	3.3	1.5	-18.4	16.7	-20.4	17.6	-22.2	18.5	-24.6	19.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-1.8	-0.7	-1.3	1.5	4.9	8.2	10.7	11.4	9.1	5.2	1.8	-0.6
	DBStd	5.61	4.44	4.03	4.27	2.97	2.40	1.81	1.64	1.79	1.97	2.70	3.25	3.86
	HDD10.0	2267	366	301	351	254	160	57	11	6	39	150	245	328
	HDD18.3	5209	624	534	609	504	418	303	236	214	277	408	495	587
	CDD10.0	100	0	0	0	0	1	4	33	49	12	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.1	7.0	7.7	7.2	7.4	6.9	6.6	6.4	6.5	6.9	7.1	7.6	7.5
Precipitation	PrecAvg	992	79	67	64	60	62	60	64	90	112	117	117	104
	PrecMax	2142	159	192	181	166	174	212	156	165	249	258	310	354
	PrecMin	594	15	11	1	5	7	3	11	28	33	37	29	26
	PrecStd	245	35	39	37	39	33	38	31	34	43	46	56	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	6.6	6.9	9.1	14.5	15.6	18.1	18.5	15.1	12.3	10.2	8.3
		MCWB	4.7	5.0	4.7	5.8	9.0	11.3	13.9	14.8	12.7	10.0	8.5	6.8
	2%	DB	5.1	5.3	5.6	7.6	11.0	13.6	15.7	16.5	13.8	11.1	8.7	6.4
		MCWB	4.1	4.2	4.0	5.0	7.5	10.1	12.7	13.9	11.8	9.5	7.6	5.4
	5%	DB	4.2	4.5	4.6	6.4	9.5	12.2	14.4	15.1	12.8	10.1	7.6	5.3
		MCWB	3.3	3.5	3.2	4.3	6.7	9.3	12.1	13.1	11.2	8.6	6.4	4.3
	10%	DB	3.4	3.7	3.7	5.5	8.3	11.1	13.3	14.0	12.1	9.2	6.4	4.3
		MCWB	2.5	2.8	2.5	3.8	6.1	8.7	11.4	12.3	10.6	7.7	5.4	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	5.6	5.6	6.7	9.5	12.0	14.6	15.5	13.9	10.8	9.0	7.2
		MCDB	5.7	6.1	6.2	8.2	13.2	14.7	17.2	17.4	14.6	11.5	9.8	8.0
	2%	WB	4.3	4.6	4.4	5.5	8.0	10.6	13.2	14.2	12.5	10.0	7.9	5.7
		MCDB	4.9	5.0	5.2	6.9	10.4	12.5	14.9	16.1	13.1	10.8	8.7	6.4
	5%	WB	3.4	3.7	3.5	4.7	7.2	9.8	12.4	13.4	11.6	8.9	6.7	4.5
		MCDB	4.1	4.2	4.3	6.0	9.0	11.6	13.9	14.8	12.4	9.9	7.5	5.2
	10%	WB	2.7	2.9	2.7	4.0	6.5	9.1	11.7	12.7	10.8	8.0	5.6	3.5
		MCDB	3.3	3.5	3.6	5.2	8.0	10.7	12.9	13.7	11.7	9.0	6.4	4.2
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.6	5.4	5.0	5.4	5.0	4.5	4.5	4.7	4.9	4.4	4.6
		MCDBR	3.9	4.5	5.4	6.3	7.5	7.1	6.6	6.1	5.6	5.4	4.8	4.3
		MCWBR	3.5	4.0	4.3	4.5	4.9	4.4	4.1	4.1	4.1	4.5	4.5	4.2
	5% WB	MCDBR	3.7	4.2	4.9	5.6	6.6	6.3	5.9	5.6	4.7	4.9	4.7	4.1
		MCWBR	3.5	4.0	4.2	4.3	4.5	4.1	4.0	4.0	3.9	4.5	4.7	4.2
Clear-Sky Solar Irradiance	taub	0.299	0.336	0.343	0.387	0.402	0.384	0.383	0.386	0.368	0.347	0.324	0.308	
	taud	2.251	2.245	2.275	2.197	2.219	2.375	2.432	2.413	2.382	2.305	2.186	2.137	
	Ebn at Noon	620	726	822	830	839	862	856	829	790	703	561	490	
	Edh at Noon	57	83	100	125	130	113	105	101	91	77	60	51	
All-Sky Solar Radiation	RadAvg	0.49	1.04	2.38	3.48	4.30	4.47	3.91	3.28	2.46	1.43	0.61	0.35	
	RadStd	0.06	0.12	0.34	0.27	0.38	0.43	0.45	0.30	0.21	0.08	0.06	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+1.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	+1.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page